

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
OPEN TEST – II
PAPER –2
TEST DATE: 20-03-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

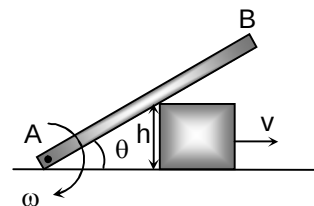
Physics

PART – I

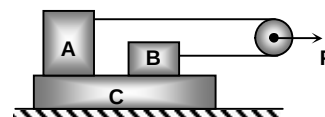
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

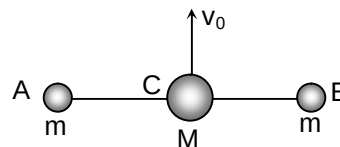
1. A thin rod AB is hinged at end A and the block is moving with a constant velocity $v = 8 \text{ m/s}$ on a horizontal surface as shown in the figure. Then choose the correct option(s). (Take $h = 1 \text{ m}$)
- (A) The angular velocity of the rod at $\theta = 30^\circ$ is 2 rad/s
 (B) The angular velocity of the rod at $\theta = 30^\circ$ is 4 rad/s .
 (C) The angular retardation of the rod at $\theta = 30^\circ$ is $4\sqrt{3} \text{ rad/s}^2$
 (D) The angular retardation of the rod at $\theta = 30^\circ$ is $8\sqrt{3} \text{ rad/s}^2$.



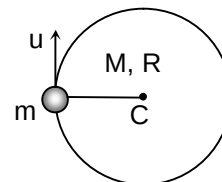
2. In the figure shown, blocks A and B of masses 4 kg and 1 kg are placed on a plank C of mass 1 kg which is placed on a smooth horizontal surface. The blocks are connected by a light inextensible string passing over an ideal pulley. The coefficients of static and kinetic friction between the blocks and the plank are 0.16 and 0.10 respectively. When a horizontal force F is applied on the pulley, the acceleration of the plank becomes 2 m/s^2 . Then choose the correct option(s). (Take $g = 10 \text{ m/s}^2$)
- (A) The magnitude of the applied force F is 18 N
 (B) The friction force acting on the block A is 1 N
 (C) The acceleration of the block A is 2 m/s^2
 (D) The acceleration of the block B is 8 m/s^2



3. A disc of mass $M = 2 \text{ kg}$ is connected with two other identical discs each of mass $m = 1 \text{ kg}$ with the help of two light inextensible threads each of length $\ell = 1 \text{ m}$. Initially the system is placed on a smooth horizontal surface and the middle disc is imparted an initial velocity $v_0 = 4 \text{ m/s}$ along the surface perpendicular to the threads as shown in the figure. When the discs A and B are about to collide, then choose the correct option(s).
- (A) The velocity of the middle disc C is 2 m/s
 (B) The velocity of the disc A is $2\sqrt{3} \text{ m/s}$
 (C) The tension in each thread is 4 N
 (D) The tension in each thread is 2 N

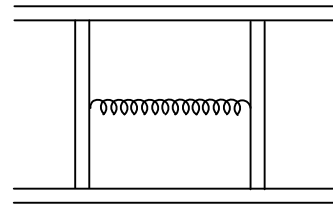


4. A small bead of mass $m = 4 \text{ kg}$ is threaded on a smooth uniform ring of mass $M = 6 \text{ kg}$ and radius $R = 0.5 \text{ m}$. Initially the system rests in a gravity free space and the bead is struck to impart it a velocity $u = 5 \text{ m/s}$ along the tangent to the ring as shown in the figure. Choose the correct option(s).



- (A) The magnitude of the force of interaction between the bead and the ring is 120 N
 (B) The magnitude of the force of interaction between the bead and the ring is 60 N
 (C) The minimum kinetic energy of the bead in subsequent motion is 26 J
 (D) The minimum kinetic energy of the bead in subsequent motion is 2 J

5. Two heat insulating pistons connected by a spring can slide without friction inside a horizontal heat insulating tube which is open at both ends. Neglect the atmospheric pressure. One mole of an ideal monoatomic gas is filled between the pistons at temperature $T_1 = 300$ K. At this temperature of the gas, the spring is not relaxed. If an amount of heat is supplied to the gas, its temperature becomes $T_2 = 400$ K and length of the spring becomes $\eta = 1.25$ times of its initial length. Then choose the correct option(s). (Take $R = 8.3$ J/mol-k)
- (A) The work done by the gas is 605.9 J
 (B) The work done by the gas is 805.9 J
 (C) The amount of heat supplied to the gas is 1850.9 J
 (D) The amount of heat supplied to the gas is 2050.9 J



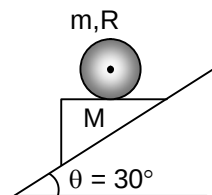
6. A source of sound with frequency $f_0 = 1700$ Hz and an observer are located at the same point. At $t = 0$, the source starts receding from the observer with a constant acceleration $a = 10$ m/s². The velocity of sound in air is $v = 340$ m/s. Then choose the correct option(s).
- (A) The frequency of sound received by the stationary observer at $t = 10$ sec is approximately 1350 Hz.
 (B) The frequency of sound received by the stationary observer at $t = 10$ sec is approximately 1550 Hz.
 (C) The sound emitted by the source at $t = 6.8$ sec (approximately) is received by the stationary observer at $t = 10$ sec.
 (D) The sound emitted by the source at $t = 8.8$ sec (approximately) is received by the stationary observer at $t = 10$ sec.

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

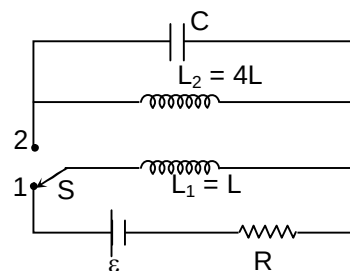
A uniform hollow cylinder of mass $m = 4$ kg and radius $R = 10$ cm rests on a wedge of mass $M = 5$ kg which is placed on a smooth inclined plane of inclination $\theta = 30^\circ$ as shown in the figure. Friction between the cylinder and the top horizontal surface of wedge is sufficient to prevent slipping of the cylinder relative to the wedge after the wedge is released from rest. (Take $g = 10$ m/s²)



7. The acceleration of the wedge is
- (A) 2 m/s²
 (B) 4 m/s²
 (C) 6 m/s²
 (D) 8 m/s²
8. The acceleration of the cylinder relative to the wedge is
- (A) $\frac{\sqrt{3}}{2}$ m/s²
 (B) $\frac{3\sqrt{3}}{2}$ m/s²
 (C) $\frac{5\sqrt{3}}{2}$ m/s²
 (D) $\frac{7\sqrt{3}}{2}$ m/s²

Paragraph for Question Nos. 09 and 10

In the circuit shown, initially the switch 'S' was getting in position-1 for a long time and then it is shifted to position-2. Initial charge on the capacitor was zero.

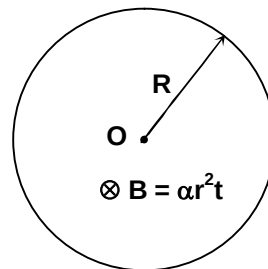


9. The maximum charge on the capacitor is
- (A) $\frac{\varepsilon}{R} \sqrt{\frac{2CL}{5}}$
- (B) $\frac{\varepsilon}{R} \sqrt{\frac{4CL}{5}}$
- (C) $\frac{\varepsilon}{R} \sqrt{\frac{6CL}{5}}$
- (D) $\frac{\varepsilon}{R} \sqrt{\frac{8CL}{5}}$
10. The maximum current through the inductor L_2 is
- (A) $\frac{8\varepsilon}{5R}$
- (B) $\frac{4\varepsilon}{5R}$
- (C) $\frac{2\varepsilon}{5R}$
- (D) $\frac{\varepsilon}{5R}$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. One end of a steel rod of length ℓ radius R and modulus of rigidity η is fixed and the other end is twisted through an angle ' θ '. If elastic deformation energy stored in the rod is $k \left(\frac{\pi \eta R^4 \theta^2}{32 \ell} \right)$. The value of k is.....
12. A time varying magnetic field $B = \alpha r^2 t$ is existing in a circular region of radius ' R ' as shown in the figure. If the induced electric field at $r = R/2$ is E_1 and at $r = 2R$ is E_2 . The ratio of $\left(\frac{E_2}{E_1} \right)$ is



13. A tube of length ℓ and radius R carries a steady flow of a liquid of density ρ and viscosity η . The velocity of flow of the liquid depends on the radial distance ' r ' from the axis of the tube as $v = v_0 \left(1 - \frac{r^2}{R^2}\right)$. If the pressure difference at the two ends of the tube is $k \left(\frac{\eta \ell v_0}{6R^2}\right)$. The value of k is.....

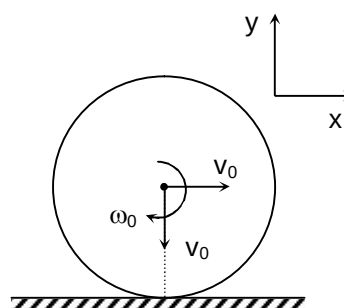
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A uniform solid sphere of mass m and radius R having initial angular velocity $\omega_0 = \frac{3v_0}{2R}$ and moving with initial velocity $(v_0\hat{i} - v_0\hat{j})$ collides with a rough horizontal surface ($\mu = 0.4$) with a coefficient of restitution $e = 0.5$ as shown in the figure. The angular velocity of the sphere just after collision becomes ω rad/s and the impulse due to friction on the sphere is J kg-m/s. (Take $m = 4.8$ kg, $R = 32$ cm and $v_0 = 3.5$ m/s)

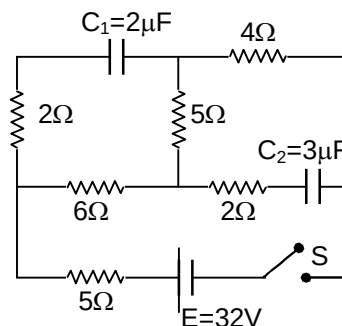


14. The value of ω is
15. The value of J is

Question Stem for Question Nos. 16 and 17

Question Stem

In the circuit shown, the switch 'S' is closed at $t = 0$. The charges on the capacitors C_1 and C_2 in the steady state are $q_1 \mu\text{C}$ and $q_2 \mu\text{C}$ respectively.

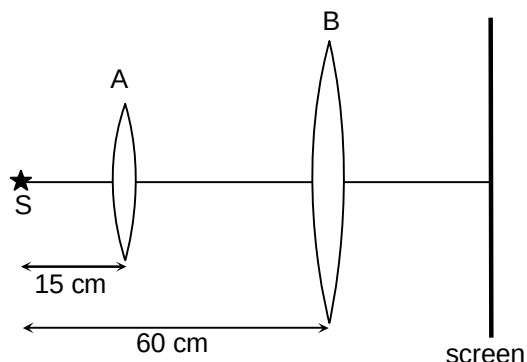


16. The magnitude of q_1 is
17. The magnitude of q_2 is

Question Stem for Question Nos. 18 and 19

Question Stem

Two lenses A and B of focal lengths 7.5 cm and 20 cm respectively are placed at distances 15 cm and 60 cm from a point source of light 'S'. The source 'S' is located on the common principal axis of the lenses. Aperture radii of the lenses A and B are 2 cm and 10 cm respectively. When a screen is placed behind the lens B at a distance d cm from the lens B, a circular light spot of minimum radius r cm is formed on the screen.



18. The value of d is

19. The value of r is

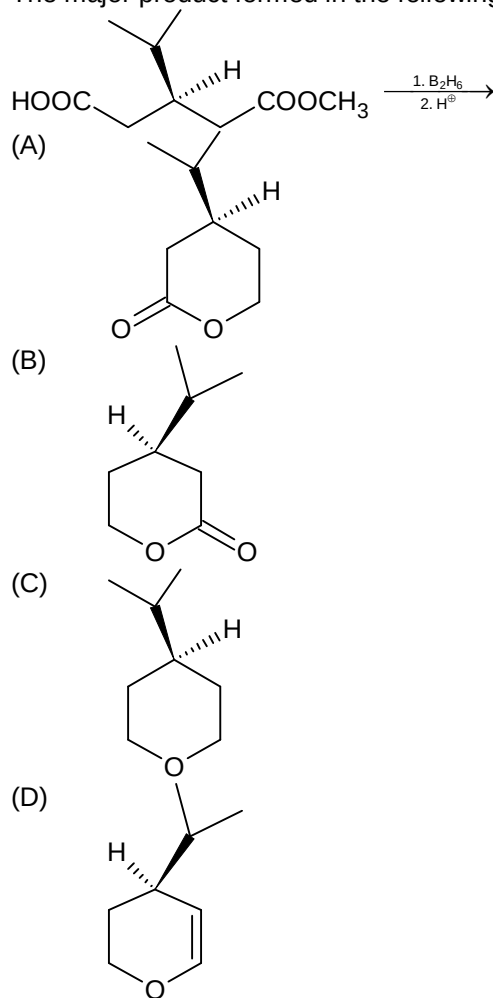
Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

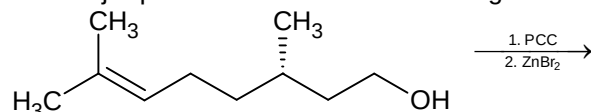
20. The major product formed in the following reaction is



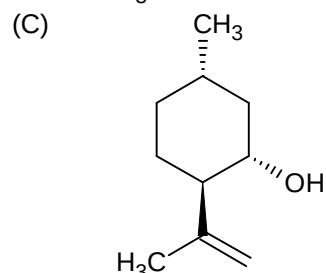
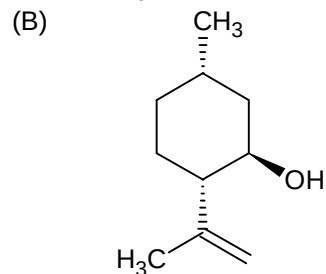
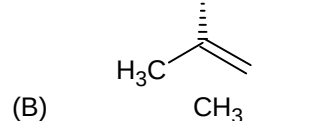
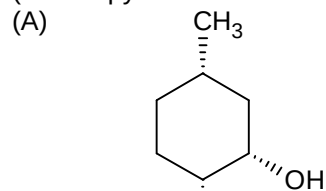
21. Which of the following statements is /are incorrect?

- (A) A Vander Waal gas at its critical condition subjected to adiabatic free expansion experiences heating effect.
- (B) If for a reaction ΔH° at 200K = ΔH° at 400K then ΔS° at 300K should be equal to ΔS° at 500K .
- (C) The value of compressibility factor of a real gas is 1 at Boyle's temperature for all range of pressures.
- (D) A Group-I metal peroxide has 68% by mass metal, hence equivalent mass of its chloride will be 79.5 gms

22. The major product formed in the following reaction is

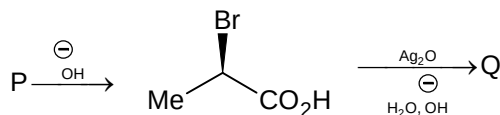


(PCC = pyridinium chlorochromate)



23. The **CORRECT** statement about the silicones is:
- (A) They are thermally unstable because of the Si–C bond.
 - (B) They are insoluble in water.
 - (C) They are organosilicon polymers.
 - (D) They have stable silica-like skeleton (–Si–O–Si–O–Si–).
24. Among the following species, the species that do not has pentagonal planar shape is
- (A) XeOF_4
 - (B) IF_5
 - (C) $[\text{SF}_5]^-$
 - (D) $[\text{XeF}_5]^-$

25. The correct statement with respect to the stereochemistry of α – hydroxy acids P and Q formed in the following reactions is:



- (A) Both P and Q are formed with retention of configuration
 (B) Both P and Q are formed with inversion of configuration
 (C) P is formed with inversion of configuration and Q with retention of configuration
 (D) P is formed with retention of configuration and Q with inversion of configuration

Section – A (Maximum Marks: 12)

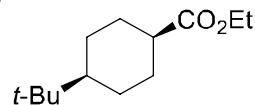
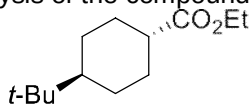
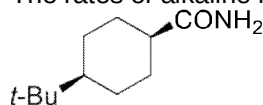
This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

Hydrolysis reactions of organic substrates are ubiquitous (common) in the environment. Hydrolysis is an important degradation reaction in surface, ground, fog and porewaters and can be a dominant pathway in biological systems as well. In general, hydrolysis occurs via one of two classes of mechanisms;

i) Nucleophilic Substitution (S_N1 and S_N2), generally occurs when the leaving group is attached to sp^3 hybridized carbon centre, such as alkyl halides, epoxides and phosphate esters. **ii) Addition – Elimination**, generally occurs when the leaving group is attached to sp^2 hybridized acyl carbon centre, such as with carboxylic acid derivatives including esters, anhydrides, amides, carbamates and ureas.

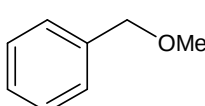
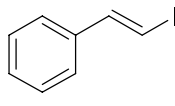
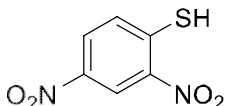
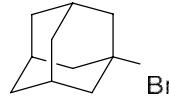
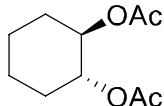
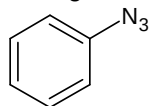
26. The rates of alkaline hydrolysis of the compounds shown below



follow the order:

- (A) I > II > III
 (B) II > I > III
 (C) II > III > I
 (D) III > I > II

27. Among the following



the compounds which can take part in nucleophilic substitution reaction are:

- (A) III, IV, and V
 (B) I, II, and VI
 (C) II, IV, and VI
 (D) I, III, and V

Paragraph for Question Nos. 28 and 29

10 mole of liquid 'X' and 10 mole of liquid 'Y' is mixed in a cylindrical vessel containing a piston arrangement. Initially a pressure of 2 atm is maintained on the solution. Now, the piston is raised slowly and isothermally. Assuming X and Y to be completely miscible and forming an ideal solution. Pure vapour pressure of X and Y is 0.6 and 0.9 atm respectively.

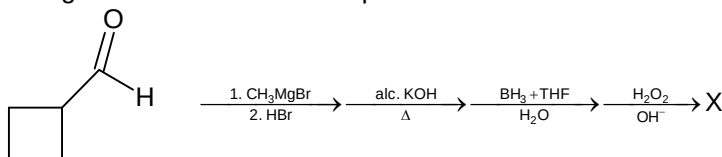
28. The minimum pressure for the existence of liquid solution is
 (A) 0.6 atm
 (B) 0.75 atm
 (C) 0.72 atm
 (D) 0.9 atm
29. If 1/4 of the total amount of liquid solution taken initially is converted to vapour, then moles of X in liquid will be
 (A) 2.125
 (B) 3.875
 (C) 7.875
 (D) 4

Section – B (Maximum Marks: 12)

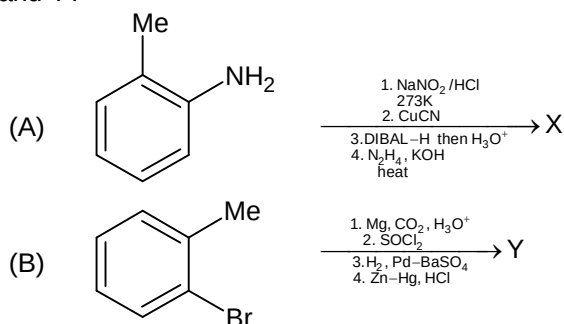
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. By using buffer concept, a student prepares 2L buffer solution of 0.330 M NaH_2PO_4 and 0.330 M Na_2HPO_4 . The solution is divided in half between the two compartments (each containing 1L buffer) of an electrolysis is carried out for 212 min with a constant current of 1.25 A. [Assumes that $\text{pK}_a(\text{H}_2\text{PO}_4^-) = 7.2$ and $\log 3 = 0.48$]
 If pH in 1st compartment cell (at anode) is x and pH in 2nd compartment cell (at cathode) is y . calculate the closest value of $10(x + y)$ -----

31. In the given reaction number of possible stereo isomers of the end major product X ?



32. The reaction sequence(s) A and B that would lead to compounds as the major product given below. calculate the total sum of DU and benzenoid position isomers for both the compounds X and Y?



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Two containers A and B of capacity 1 litre and 2 litre respectively is connected by tube of negligible volume. The tube is initially closed by stop cock and 'A' contains small amount of $\text{H}_2\text{O}(\text{l})$ and B contain initially pure $\text{COCl}_2(\text{g})$ at partial pressure of 100 torr. The gas COCl_2 dissociates at experimental condition into $\text{CO}(\text{g})$ and $\text{Cl}_2(\text{g})$. Now the stopcock connecting two container is opened and sufficient time is given to attain final equilibrium. The final pressure at equilibrium is found to be 100 torr in both containers. Using given data and taking following assumption if we have calculated the the mole fraction of $\text{CO}(\text{g})$ in final state is 'x' and the final partial pressure of $\text{COCl}_2(\text{g})$ is 'y'.

Given : Vapour pressure of H_2O at experimental temperature 'T' is equal to 20 torr.

Assumption-1 : Volume of $\text{H}_2\text{O}(\text{l})$ small and can be neglected but sufficient to create vapour pressure in both container.

Assumption-2 : COCl_2 , CO and Cl_2 gas do not dissolve in $\text{H}_2\text{O}(\text{l})$.

Assumption-3 : Whole experiment is carried out under isothermal condition.

33. The value of x is _____

34. The value of y is _____

Question Stem for Question Nos. 35 and 36

Question Stem

For the following reaction $2\text{X} + \text{Y} \xrightarrow{k} \text{P}$ the rate of reaction is $\frac{d(\text{P})}{dt} = k[\text{X}]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution. At 100 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is(are) (Use: $\ln 2 = 0.693$)

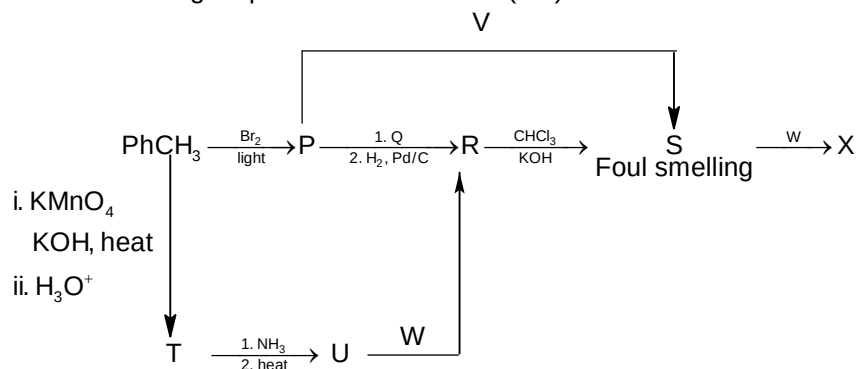
35. The rate constant, k, (in 10^{-4} s^{-1}) of the reaction is ____.

36. At 200 s, $-\frac{d[\text{Y}]}{dt}$ in (in $10^{-3} \text{ mol L s}^{-1}$) is ____.

Question Stem for Question Nos. 37 and 38

Question Stem

For the following sequence of reactions is (are)



If molar mass of the metal in compound Q and V is 107.8 g/mol.

37. The value of sum of molar mass of compound (Q,V and R) is ____.
38. The value of TOTAL number of benzene ring containing structural isomers of the compound X is ____.

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let $f : [0, 2] \rightarrow [0, 2]$ be continuous and differentiable function, does not have local maximum or minimum such that $f(f(f(x))) + f(x) = 2 \forall x \in [0, 2]$, then
 (A) $f(x)$ is increasing function $\forall x \in [0, 2]$
 (B) $f(x)$ is decreasing function $\forall x \in [0, 2]$
 (C) $f(x)$ is non increasing function $\forall x \in [0, 2]$
 (D) $\int_0^2 f(x) dx = 2$
40. If $f : [0, 1] \rightarrow [0, 1]$ is an onto function such that $f'(x) > 0$ and $f''(x) < 0$, then
 (A) $\frac{f(x)}{x} \forall x \in (0, 1]$ is decreasing function
 (B) $\frac{f(x)}{x} \forall x \in (0, 1]$ is increasing function
 (C) $f(x) \cdot f^{-1}(x) \leq x^2$ for some $x \in (0, 1)$
 (D) $f^{-1}(x) \leq x \leq f(x), x \in [0, 1]$
41. Let $a_1, a_2, a_3, \dots, a_n$ be a sequence such that $a_n = 2a_{n-1} - 1, a_1 = 3$ for $n \in \mathbb{N}$, then
 (A) $a_1, a_2, a_3, \dots, a_n$ are in Arithmetic Progression
 (B) $a_n > 2021$ for $n \geq 11$
 (C) $a_1 - 1, a_2 - 1, a_3 - 1, \dots, a_n - 1$ are in Geometric Progression
 (D) none of these
42. Let A be $n \times n$ matrix, $A = [a_{ij}]_{n \times n}$ such that $a_{ij} = \begin{cases} n & ; \text{ if } i = j \\ -1 & ; \text{ if } i \neq j \end{cases}$, then
 (A) $A^{-1} = [b_{ij}]_{n \times n}, b_{ij} = \begin{cases} \frac{n}{n+1} & ; i = j \\ n & ; i \neq j \end{cases}$
 (B) $A^{-1} = [b_{ij}]_{n \times n}, b_{ij} = \begin{cases} \frac{2}{n+1} & ; i = j \\ \frac{1}{n+1} & ; i \neq j \end{cases}$
 (C) $A^{-1} = [b_{ij}]_{n \times n}, b_{ij} = \frac{n}{n+1}, \forall i, j$
 (D) $A^{-1} = [b_{ij}]_{n \times n}, b_{ij} = \frac{2}{n+1}, \forall i, j$
43. Let n_1 and n_2 be two three digit numbers of form $n_1 = x_1x_2x_3, n_2 = y_1y_2y_3$ where $1 \leq x_3 < x_2 < x_1 \leq 9$ and $1 \leq y_3 < y_2 < y_1 \leq 8$, then probability that $n_1 \geq n_2$ is
 (A) $\frac{37}{56}$
 (B) $\frac{111}{{}^9C_3 \cdot {}^8C_3}$

- (C) $\frac{113}{168}$
 (D) none of these
44. Let T is a parallopiped in which A, B, C, D are vertices of one face on the plane $2x - 2y + z = 0$ and if face above it has corresponding vertices A', B', C' and D' lie on plane $2x - 2y + z + 4 = 0$. If T is now compressed to 'S' with face A, B, C, D remaining same and A', B', C', D' shifting to A'', B'', C'' and D'' in 'S' and volume of parallopiped S is reduced to 75% of T, then
- (A) there exist infinite point A'' on line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$
 (B) A'' lie on the plane $2x - 2y + z + 3 = 0$
 (C) there exist infinite point A'' on the line $\frac{x+1}{3} = \frac{y+1}{2} = z$
 (D) there exist exactly two distinct points A'' on the line $\frac{x+1}{3} = \frac{y+1}{2} = z$

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 45 and 46

Let AD and CF are mediums from vertex A and C respectively, of triangle ABC, and G be the centroid of triangle. Four points B, D, G and F are concyclic

45. If angle $C \in \left(\frac{2\pi}{3}, \pi\right)$, then
- (A) $\frac{a}{b} \in \left(\frac{\sqrt{3}+1}{2}, 1\right)$
 (B) $\frac{a}{b} \in \left(\frac{\sqrt{3}-1}{2}, 1\right)$
 (C) $\frac{a}{b} \in \left(\frac{\sqrt{3}-1}{2}, \frac{1}{2}\right)$
 (D) $\frac{a}{b} \in \left(\frac{1}{2}, \frac{\sqrt{3}+1}{2}\right)$
46. If angle $C \in \frac{2\pi}{3}$ and $AB = 7$ units, then
- (A) $b = 2\sqrt{7}$
 (B) $a = 2\sqrt{7}$
 (C) $b = \sqrt{7}$
 (D) none of these

Paragraph for Question Nos. 47 and 48

Seven friends $F_1, F_2, F_3, F_4, F_5, F_6$ and F_7 go to a restaurant and keep their hats and coats on stand at entrance, while going back they pickup one coat and hat at random. Now answer the below question

47. Probability that no one get his own coat given that exactly F_1 and F_2 has got their own hat

- (A) $\frac{309}{6!}$
 (B) $\frac{1854}{7!}$
 (C) $\frac{309}{7!}$
 (D) none of these

48. Probability that each get their own hat but no one get their own coat

- (A) $\frac{1854}{7!}$
 (B) $\frac{1857}{6!}$
 (C) $\frac{1854}{(7!)^2}$
 (D) none of these

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. If the area of a square circumscribing an ellipse is A square units. The maximum distance of a normal from the centre of ellipse is r , then the value of $\left[\frac{A}{r^2} \right]$ is

(Where $[.]$ denotes greatest integer function) (Given eccentricity of ellipse is 0.5)

50. The number of solution of equation $\cos z = i \sin(z)$, where $z \in \mathbb{C}$ is

51. If $a, b, c \in \mathbb{N}$, the probability that $a^2 + b^2 + c^2 = 13k$, $k \in \mathbb{N}$ is $\frac{m}{n}$, where m, n are relatively prime numbers, then $(n - m)$ is

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $f : [-1, 5] \rightarrow \mathbb{R}$ be a function, such that $f(x) = (x - 2n)^2$, $2n - 1 \leq x \leq 2n + 1$, $n = 0, 1, 2$ and $P(x) = \min\{f(t)\}$, $l \leq t \leq x$, $l < x \leq l + 1$ where $l = -1, 0, 1, 2, 3, 4$ and $g(x) = f(x) - p(x) \forall x \in [-1, 5]$

52. If $I = \int_{-1}^5 f(x) dx$ and $J = \int_{-1}^5 g(x) dx$, then $\frac{I}{J}$ is

53. $y = f(x)$ is non-differentiable at α number of points and $y = g(x)$ is discontinuous at β number of points for $x \in (-1, 5)$, then $\alpha + \beta$ is

Question Stem for Question Nos. 54 and 55

Question Stem

Let $\sum_{r=1}^{\infty} \frac{1}{r^2} = a$ and $f(x) = \frac{\ln x}{1-x}$, $g(x) = \left(\frac{x \ln x}{1-x^2} \right)$

54. Let $I_1 = \int_0^1 f(x) dx$ and $I_2 = \int_0^1 g(x) dx$, then $\frac{|(I_1 - I_2)|}{a}$ is

55. If $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r(\ln r - \ln n)}{n^2 - r^2} \right) = \lambda a$, then the value of $|\lambda|$ is

Question Stem for Question Nos. 56 and 57

Question Stem

Let the median AD of $\triangle ABC$ intersects the incircle of triangle at K and L (such that K is nearer to vertex A as compare to L). If $AK = KL = LD$, and semi-perimeter of $\triangle ABC$ is $s = 14$ units, then

56. Diameter of circumcircle of $\triangle ABC$ is

57. Radius of in-circle of the triangle ABC is